Microprice as a Centered Poisson Corrector

Paper I: move-time structure, centering, and horizon effects

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Abstract. This paper has two goals. First, it identifies the object that classical microprice estimates after the event-time model is reduced to price-move time. For a finite top-of-book Markov model with no-move kernel Q, move kernel T, event-time one-step drift

$$r(x) := \mathbb{E}[p_{n+1} - p_n \mid X_n = x],$$

and fundamental matrix $N = (I - Q)^{-1}$, define

$$G_1 := Nr, \qquad B := NT.$$

The n-move predictor is then

$$G^{(n)} = \sum_{j=0}^{n-1} B^j G_1.$$

Second, it explains why the long-run correction can fail at short horizons even when the move-time model is well specified. If the move-time chain is ergodic with stationary distribution π, mean move drift $\mu = \pi^\top G_1$, and projector $\Pi_B := \mathbf{1}\pi^\top$, then

$$G^{(n)} = n\mu\,\mathbf{1} + g - B^n g, \qquad (I - B)g = G_1 - \mu\mathbf{1}, \qquad \pi^\top g = 0,$$

and the long-run move-time microprice level is

$$G^* := (I - B + \Pi_B)^{-1} G_1 = g + \mu\mathbf{1}.$$

Thus classical microprice is a centered Poisson corrector on the move-time chain, while the transient term $-B^n g$ governs the horizon effect. We also study the move-time discounted resolvent $(I - \beta B)^{-1} G_1$, where β discounts across price moves after the reduction; this is distinct from the event-time discount γ used in the companion note *Microprice, Bellman Values, and Semi-Markov Reinforcement Learning*.

Empirically, this decomposition predicts a horizon effect: at short horizons the transient term can still dominate, so G_1 or even the raw mid can outperform G^*; at longer horizons the centered correction wins when the move-time martingale diagnostic remains small. The public BAC/CVX benchmarks are mixed, WSELOB shows the clearest crossover, and a one-day IEX panel isolates the residual-drift failure mode.

1 Introduction

Stoikov's microprice [1] propagates expected future price changes through an absorbing top-of-book Markov chain. This paper is the structural half of a two-note sequence. It identifies the long-run object computed by the classical construction and explains why that object need not be the best short-horizon forecast.

Throughout both notes we use the same dictionary:

$$r(x) := \mathbb{E}[p_{n+1} - p_n \mid X_n = x], \qquad N := (I - Q)^{-1}, \qquad G_1 := Nr, \qquad B := NT.$$

If the move-time chain B is ergodic with stationary distribution π, we write

$$\mu := \pi^\top G_1, \qquad \Pi_B := \mathbf{1}\pi^\top, \qquad (I - B)g = G_1 - \mu\mathbf{1}, \qquad \pi^\top g = 0,$$

and

$$G^* := (I - B + \Pi_B)^{-1} G_1 = g + \mu\mathbf{1}.$$

Here g is the centered move-time relative value and G^* the uncentered long-run move-time microprice level; in centered regimes, $G^* = g$.

Paper I analyzes the move-time Poisson decomposition, the martingale correction, and the horizon effect. Paper II keeps the same objects (r, N, G_1, B, g, G^*) and adds exact event-time Bellman targets together with move-time TD/SMDP learning rules. Empirically, the public BAC/CVX benchmarks are mixed, WSELOB gives the clearest horizon crossover, and the one-day IEX panel isolates the residual-drift failure mode.

2 From event time to move time

Let X_n be a finite-state Markov approximation of the top of book at event time, and let p_n be the mid-price. Decompose the transition kernel according to whether the mid-price moves:

$$P = Q + T,$$

where Q_{xy} is the probability of moving from state x to state y without changing the mid-price and T_{xy} is the probability of moving from x to y with a price change. Define the event-time one-step drift

$$r(x) := \mathbb{E}[p_{n+1} - p_n \mid X_n = x].$$

Assume $\rho(Q) < 1$, so the no-move subchain is transient and

$$N := (I - Q)^{-1} = \sum_{k \geq 0} Q^k$$

exists. Set

$$G_1 := Nr, \qquad B := NT.$$

Proposition 2.1 (Embedded move-time chain)**.** *Let*

$$\tau_0 := 0, \qquad \tau_{k+1} := \inf\{n > \tau_k : p_n \neq p_{n-1}\}, \qquad Y_k := X_{\tau_k}.$$

Then Y_k is a Markov chain with transition kernel B. Moreover,

$$G_1(x) = \mathbb{E}_x[p_{\tau_1} - p_{\tau_0}],$$

so G_1 is the expected price change at the next move conditional on the current state.

Thus the event-time recursion collapses to move time: B is the price-move kernel and G_1 the one-move drift. The companion note uses the discounted counterparts

$$G_{1,\gamma} := (I - \gamma Q)^{-1} r, \qquad B_\gamma := \gamma (I - \gamma Q)^{-1} T,$$

which discount each no-move excursion before it is collapsed. In this paper, β is reserved for the move-time resolvent $(I - \beta B)^{-1} G_1$, which discounts only across already-collapsed price moves.

If π denotes the stationary distribution of B, define

$$\mu := \pi^\top G_1, \qquad \Pi_B := \mathbf{1}\pi^\top.$$

Remark 2.2 (Centering)**.** *When B is ergodic, $B^n \to \Pi_B$. Hence Stoikov's convergence condition $B^n G_1 \to 0$ is equivalent to*

$$\pi^\top G_1 = 0.$$

Only the invariant constant mode can obstruct convergence.

3 The centered Poisson corrector

Define the n-move price adjustment by

$$G^{(n)} := \sum_{j=0}^{n-1} B^j G_1.$$

Equivalently,

$$\mathbb{E}[p_{\tau_n} - p_{\tau_0} \mid Y_0 = x] = G^{(n)}(x).$$

Proposition 3.1 (Finite-horizon decomposition)**.** *Assume that B is ergodic with stationary distribution π, and set*

$$h := G_1 - \mu\mathbf{1}.$$

Let g be the unique centered solution of

$$(I - B)g = h, \qquad \pi^\top g = 0.$$

Then, for every $n \geq 1$,

$$G^{(n)} = n\mu\,\mathbf{1} + g - B^n g.$$

If

$$G^* := (I - B + \Pi_B)^{-1} G_1,$$

then

$$G^* = g + \mu\mathbf{1}.$$

In particular, if $\mu = 0$, then $G^* = g$ and

$$G^{(n)} \to G^*.$$

Thus the predictor splits into linear drift, a bounded state correction, and the transient remainder $-B^n g$. The empirical horizon effect is entirely that last term.

Corollary 3.2 (Move-time martingale correction)**.** *Under the assumptions of Proposition 3.1,*

$$M_k := p_{\tau_k} - k\mu + g(Y_k)$$

is a martingale with respect to the move-time filtration. When $\mu = 0$, the corrected price

$$p_{\tau_k} + G^*(Y_k)$$

is a move-time martingale.

Proposition 3.3 (Move-time discounted regularization)**.** *For $\beta \in (0,1)$ define*

$$g_\beta := (I - \beta B)^{-1} G_1.$$

Then

$$g_\beta = \frac{\mu}{1-\beta}\mathbf{1} + (I - \beta B)^{-1}(G_1 - \mu\mathbf{1}).$$

Hence the only singular term as $\beta \uparrow 1$ is the constant mode. If $\mu = 0$ and, for some $\rho \in (0,1)$,

$$\|B^k G_1\|_\infty \le C_\rho \rho^k \|G_1\|_\infty \qquad \text{for all } k \ge 0,$$

then

$$\|g_\beta - G^*\|_\infty \le (1-\beta)\frac{C_\rho \rho}{(1-\rho)^2}\|G_1\|_\infty.$$

The resolvent is therefore best viewed as a regularized proxy for G^*: nearly lossless when $\mu \approx 0$, but dominated by the constant mode when centering fails. Unlike $V_\gamma = (I - \gamma P)^{-1} r$ in Paper II, it discounts only after the move-time reduction.

4 Empirical results

4.1 Evaluation protocol

We evaluate predictors against grouped state-conditioned future-price curves rather than pointwise forecast errors. For a horizon h, let c index a spread/imbalance bucket. Write $\bar{\Delta}_h(c)$ for the average future mid-price change over all rows in bucket c, and write $\bar{f}(c)$ for the average predictor value in the same bucket. The reported score is

$$\mathrm{RMSE}_h(f) := \left(\frac{1}{|C_h|}\sum_{c \in C_h}\left(\bar{\Delta}_h(c) - \bar{f}(c)\right)^2\right)^{1/2}.$$

The horizons are fixed clock/event-time horizons rather than exact move counts, so the empirical question is qualitative: does the long-run correction become more useful as the horizon lengthens? Throughout this paper, "Disc(.99)" means the move-time resolvent $(I - 0.99\,B)^{-1} G_1$; the companion note reserves $V_{0.99}$ for event-time discounting on P.

To connect the data back to Corollary 3.2, we also report the move-time martingale diagnostic. Writing

$$\Delta P_k := p_{\tau_{k+1}} - p_{\tau_k}, \qquad \widehat{\xi}_k := \Delta P_k - \widehat{\mu} + G^*(Y_{k+1}) - G^*(Y_k), \qquad \widehat{t}_M := \left|\frac{\overline{\widehat{\xi}}}{s_{\widehat{\xi}}/\sqrt{n}}\right|.$$

Small values of $\widehat{t}_M$ indicate that the estimated correction is close to a move-time martingale.

4.2 Public BAC/CVX benchmarks

Table 1 and Figure 1 summarize the public BAC/CVX benchmarks. BAC is the mixed case: G^* materially improves on the raw mid but not on the weighted mid. CVX is the cleaner public success case for G^*. In all three benchmark cells, $\widehat{t}_M < 2$.

Table 1. Public BAC/CVX grouped-RMSE benchmarks and move-time diagnostics.

Asset	Horizon (s)	Raw mid	Weighted mid	G_1	G^*	Disc(.99)	$\widehat{t}_M$
BAC	60	0.001812	0.000435	0.000909	0.000892	0.000892	1.12
BAC	180	0.002287	0.000345	0.000561	0.000547	0.000547	1.12
CVX	10	0.001828	0.003941	0.000910	0.000697	0.000699	0.84

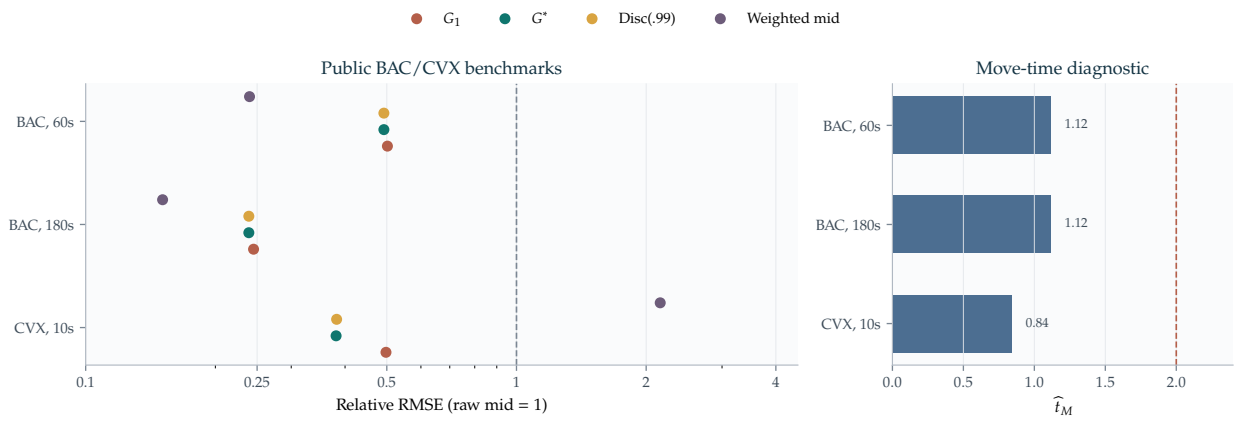

Figure 1. Public BAC/CVX benchmarks. Left: grouped RMSE relative to the raw mid. Right: the move-time martingale diagnostic. BAC remains a nuanced benchmark in which the weighted mid is highly competitive, whereas CVX is the cleaner public success case for G^*.

4.3 WSELOB: the horizon crossover

WSELOB gives the clearest horizon crossover. At 10 seconds, the raw mid is best on PEKAO, PKOBP, and PZU, while G_1 is best on KGHM and PKNORLEN; G^* can still be between 6% and 83% worse than the raw mid, consistent with a material transient. By 60 seconds the ordering reverses: G^* is best on four names and Disc(.99) on the fifth, with grouped-RMSE reductions of 20% to 56% relative to the raw mid. The weighted mid is not competitive, and $\widehat{t}_M < 2$ for every name.

Table 2. WSELOB summary by asset.

Asset	Best @ 10s	Best @ 60s	Raw/G^* @ 60s	WMid/G^* @ 60s	$\widehat{t}_M$
KGHM	G_1	G^*	1.25	2.07	1.97
PEKAO	Raw mid	Disc(.99)	1.97	6.61	1.08
PKNORLEN	G_1	G^*	1.54	4.14	0.68
PKOBP	Raw mid	G^*	2.03	4.94	1.85
PZU	Raw mid	G^*	2.26	6.37	0.77

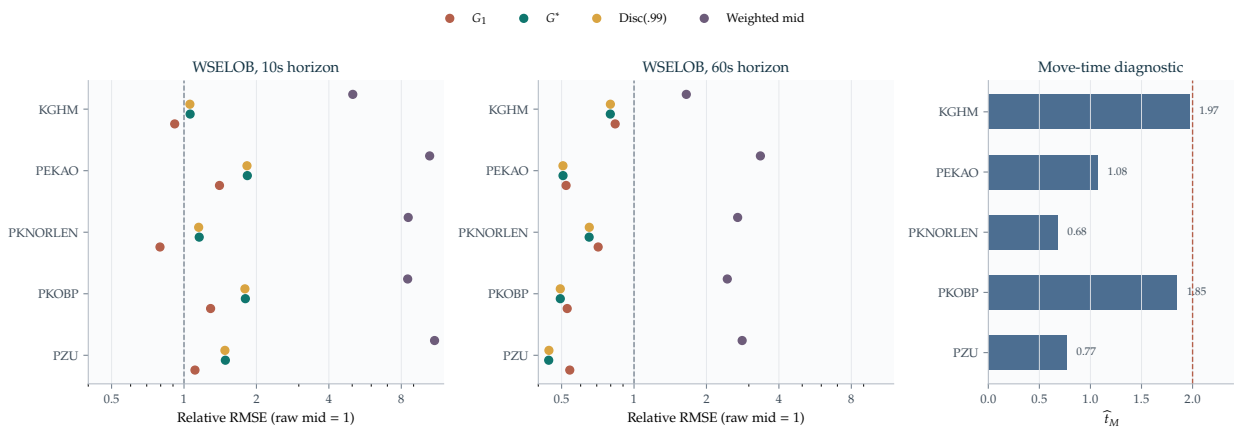

Figure 2. WSELOB summary. Relative RMSE is normalized by the raw mid, so values below one indicate an improvement over the raw mid. The crossover is clear: the limiting correction is not yet best at 10 seconds, but it becomes best by 60 seconds once the horizon is long enough.

4.4 IEX: centering as a diagnostic

The IEX venue-local panel is deliberately harder: a single venue, a short sample, and a coarse state description leave more room for residual drift. BAC remains near-centered with $\widehat{t}_M = 0.16$ and G^* modestly improves on the raw and weighted mid. SPY and GLD do not: $\widehat{t}_M = 3.09$ and 2.24, and the discounted surrogate no longer shadows the centered correction. This is the failure mode of Proposition 3.3: when the constant mode is not well centered, discounted extrapolation is driven by residual drift rather than by the bounded state correction.

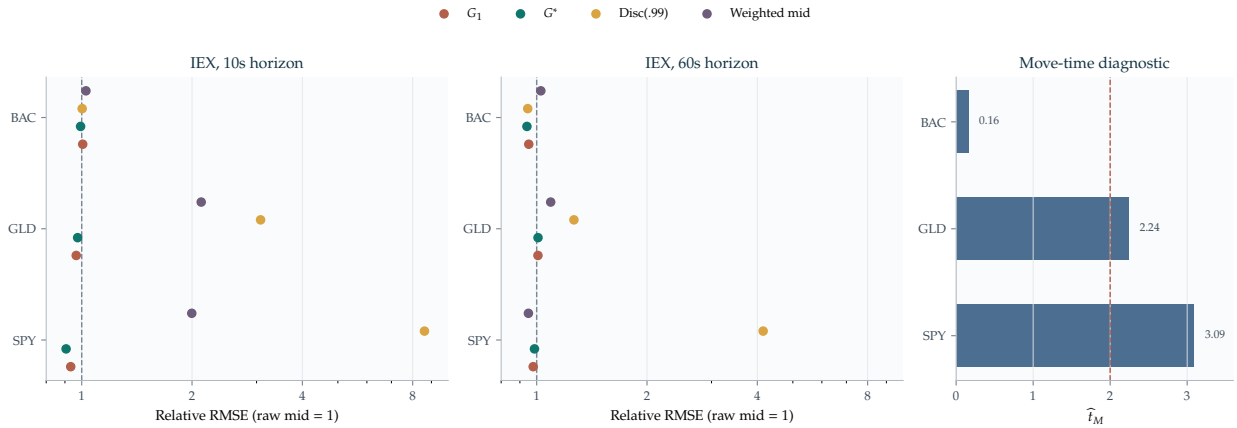

Figure 3. IEX venue-local stress test. BAC remains near-centered, while SPY and GLD violate the move-time martingale diagnostic. In those names the discounted resolvent no longer behaves like a harmless regularization of G^*.

5 Conclusion

Sampling at move times turns microprice into a finite-state Poisson problem. The event-time recursion collapses to a move-time kernel B and a one-move drift G_1, and the n-move predictor decomposes as

$$G^{(n)} = n\mu\mathbf{1} + g - B^n g.$$

This identity isolates the martingale correction and the horizon effect.

Empirically, the correction works best when the horizon is long enough for $B^n g$ to decay and the move-time martingale diagnostic remains small. WSELOB is the clearest positive case, BAC/CVX are useful public calibrations, and the IEX panel shows the residual-drift failure mode. Paper II starts from the same primitives (r, N, G_1, B, g, G^*) and adds exact event-time Bellman objectives together with move-time TD/SMDP learning rules.

A Proofs

Proof of Proposition 2.1. The first price move occurs after n no-move steps and one move step with kernel $Q^n T$. Summing over $n \geq 0$ gives

$$\mathbb{P}_x(Y_1 = y) = \sum_{n\geq 0}(Q^n T)_{xy} = ((I-Q)^{-1}T)_{xy} = B_{xy}.$$

Thus B is the move-time transition kernel. For the reward identity,

$$\mathbb{E}_x[p_{\tau_1} - p_{\tau_0}] = \sum_{n\geq 0}(Q^n r)_x = ((I-Q)^{-1}r)_x = G_1(x).$$

The Markov property of (Y_k) follows from the strong Markov property at the stopping times (τ_k). □

Proof of Proposition 3.1. Write $G_1 = \mu\mathbf{1} + h$ with $\pi^\top h = 0$. Since $h = (I-B)g$,

$$G^{(n)} = \sum_{j=0}^{n-1} B^j G_1 = n\mu\mathbf{1} + \sum_{j=0}^{n-1} B^j (I-B)g = n\mu\mathbf{1} + g - B^n g.$$

Also,

$$(I - B + \Pi_B)(g + \mu\mathbf{1}) = (I-B)g + \Pi_B g + \mu\Pi_B\mathbf{1} = h + \mu\mathbf{1} = G_1,$$

so $G^* = g + \mu\mathbf{1}$. If $\mu = 0$, then $G^* = g$ and ergodicity gives $B^n g \to \Pi_B g = \mathbf{1}\pi^\top g = 0$, hence $G^{(n)} \to G^*$. □

Proof of Corollary 3.2. By Proposition 3.1,

$$(I-B)g = G_1 - \mu\mathbf{1}.$$

Hence

$$\mathbb{E}[p_{\tau_{k+1}} - p_{\tau_k} \mid Y_k] = G_1(Y_k),$$

while

$$\mathbb{E}[g(Y_{k+1}) - g(Y_k) \mid Y_k] = (Bg - g)(Y_k) = -(G_1(Y_k) - \mu).$$

Adding the two conditional expectations yields

$$\mathbb{E}[M_{k+1} - M_k \mid Y_k] = 0.$$

□

Proof of Proposition 3.3. The Neumann series gives

$$g_\beta = \sum_{k\geq 0} \beta^k B^k G_1.$$

With $G_1 = \mu\mathbf{1} + h$ and $B^k\mathbf{1} = \mathbf{1}$,

$$g_\beta = \sum_{k\geq 0} \beta^k(\mu\mathbf{1} + B^k h) = \frac{\mu}{1-\beta}\mathbf{1} + (I - \beta B)^{-1}h.$$

If $\mu = 0$, then $h = G_1$ and

$$G^* - g_\beta = \sum_{k\geq 0}(1-\beta^k)B^k G_1.$$

The $k = 0$ term vanishes, and for $k \geq 1$ we have $1 - \beta^k \leq k(1-\beta)$. Hence

$$\|G^* - g_\beta\|_\infty \leq (1-\beta)C_\rho\|G_1\|_\infty \sum_{k\geq 1} k\rho^k = (1-\beta)\frac{C_\rho\rho}{(1-\rho)^2}\|G_1\|_\infty.$$

□

References

[1] S. Stoikov, *The Micro-Price: A High Frequency Estimator of Future Prices*, Quantitative Finance, 18(12):1959–1966, 2018.